

TRIVANDRUM INTL, India

WMO: 433720

Lat: 8.482N

Lon: 76.920E

Elev: 15

StdP: 14.69

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	73.2	74.8	62.8	85.5	79.4	65.8	95.2	79.9	18.0	83.1	16.6	84.0	2.3	120	0.368

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.5	93.0	81.7	91.7	80.9	91.2	80.6	82.9	89.9	82.3	89.2	81.7	88.5	10.0	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.9	160.6	87.5	80.6	158.7	87.5	79.2	151.7	86.5	46.7	90.4	46.0	89.8	45.4	89.4	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.5	13.4	12.0	DB	70.3	95.0	1.3	1.9	69.3	96.4	68.6	97.5	67.9	98.6	67.0	100.0	(4)
(5)			WB	64.6	84.8	3.0	1.1	62.5	85.6	60.7	86.2	59.0	86.8	56.8	87.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.9	81.5	82.9	84.9	86.2	84.5	82.3	81.6	81.6	81.9	82.5	82.5	82.3
	DBStd	2.19	1.60	1.76	1.67	1.82	2.24	1.87	1.67	1.66	1.39	1.28	1.35	1.26
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12005	975	921	1082	1085	1069	968	979	981	958	1009	975	1003
	CDD65	6530	510	501	617	635	604	518	514	516	508	544	525	538
	CDH74	75995	5849	6175	8098	8670	7687	5611	5228	5200	5453	6048	5868	6108
	CDH80	29141	2238	2599	3817	4418	3448	1765	1470	1434	1632	2031	2052	2237
Wind	WSAvg	5.4	3.5	4.0	4.6	5.8	6.6	6.4	7.7	7.7	6.5	4.7	3.4	3.5
Precipitation	PrecAvg	70.1	0.7	0.9	1.5	4.7	8.0	11.9	8.0	5.9	7.1	10.9	8.1	2.5
	PrecMax	95.3	6.4	5.0	12.3	12.2	23.3	34.3	16.6	20.7	19.6	25.8	28.4	13.5
	PrecMin	44.4	0.0	0.0	0.0	0.2	0.3	2.3	1.9	0.8	0.2	0.7	0.6	0.0
	PrecStd	12.6	1.2	1.3	1.9	2.7	5.0	6.6	3.5	3.9	5.4	5.7	5.1	2.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.0	91.7	93.5	94.9	92.9	90.0	89.7	89.7	89.6	89.9	90.0	90.0
		MCWB	77.6	79.0	80.4	82.8	82.1	80.4	80.2	80.0	80.2	79.2	79.4	77.9
	2%	DB	89.6	91.0	91.8	93.3	91.6	89.3	88.1	88.1	88.1	89.2	89.5	89.4
		MCWB	77.9	78.8	80.2	82.3	81.5	80.1	80.0	79.9	79.8	79.2	79.3	78.3
	5%	DB	88.1	89.7	91.4	91.8	91.1	87.9	87.5	87.5	87.6	88.0	88.1	88.1
		MCWB	77.3	78.0	80.0	81.6	81.2	80.2	79.9	79.8	79.7	79.5	79.4	78.6
	10%	DB	87.8	89.2	90.0	91.3	89.7	86.4	86.0	86.0	86.2	87.5	87.7	87.7
		MCWB	77.2	77.7	79.3	81.4	80.4	79.5	79.1	78.9	79.1	79.4	79.4	78.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.3	81.3	82.8	84.2	83.6	82.3	81.9	81.1	82.5	81.9	81.8	81.5
		MCDB	87.6	89.4	90.9	92.5	90.0	86.4	87.0	87.2	85.4	86.1	86.5	86.9
	2%	WB	79.8	80.1	81.7	83.2	82.7	81.1	80.7	80.1	80.8	80.8	80.7	80.5
		MCDB	87.0	88.2	89.7	91.4	89.4	86.2	86.7	86.2	86.0	86.5	87.1	86.7
	5%	WB	79.1	79.6	80.9	82.5	81.8	80.5	80.0	79.8	80.0	80.1	80.0	79.9
		MCDB	86.3	87.7	89.0	90.6	88.5	85.9	85.8	85.6	85.4	86.0	86.4	86.1
	10%	WB	78.4	78.9	80.2	81.8	81.1	79.9	79.3	79.2	79.5	79.7	79.7	79.4
		MCDB	85.6	87.0	88.1	89.7	87.7	85.1	84.9	84.8	84.8	85.5	85.9	85.6
Mean Daily Temperature Range	5% DB	MDBR	12.9	12.6	11.7	10.5	9.9	8.7	9.0	9.1	9.0	9.7	10.4	11.2
		MCDBR	13.5	13.1	12.6	11.0	10.6	10.0	10.4	10.6	10.2	10.7	11.3	11.9
		MCWBR	7.0	6.8	5.7	5.4	4.7	5.0	5.9	5.7	5.0	5.2	5.3	5.9
	5% WB	MCDBR	12.3	11.9	11.5	10.8	9.9	9.1	9.8	9.9	9.5	10.0	10.6	10.9
		MCWBR	6.7	6.6	5.4	5.7	5.1	5.0	6.1	5.9	5.3	5.4	5.4	5.8
Clear-Sky Solar Irradiance	taub	0.476	0.500	0.569	0.577	0.577	0.531	0.550	0.526	0.483	0.517	0.527	0.497	
	taud	2.202	2.129	1.960	1.962	1.970	2.107	2.035	2.117	2.256	2.147	2.097	2.164	
	Ebn at Noon	266	263	246	241	236	244	240	250	265	255	249	257	
	Edh at Noon	46	51	61	60	58	50	54	51	45	49	51	47	
All-Sky Solar Radiation	RadAvg	1736	1950	2009	1895	1726	1586	1624	1715	1786	1624	1408	1519	
	RadStd	80	76	117	93	138	130	107	120	124	109	90	96	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	+0.50	N/A	N/A	+0.99	+1.08	+0.99	N/A	N/A	N/A	N/A

Nomenclature: See separate page